Alejandro Aristizábal Sánchez

Tech Lead | AI & Machine Learning | Software Engineering | MSc Artificial Intelligence

alejandro.a01@outlook.com

GitHub | LinkedIn

Summary

Tech Lead with 7+ years shipping ML/AI products end to end and owning their outcomes across travel, CPG and energy. Led a personalization platform from pilot to fleet adoption, generating USD 5M+ and 10M+ recommendations, while cutting run cost by 30% with p95 < 150ms.

Previously co-built a global ML platform used in 8 countries, delivering USD 100M+ impact and 70%+ quality (ML Assessment) gains. Earlier roles in teaching/backend and analytics grounded my approach to reliability and delivery.

Operate across Azure and GCP/AWS, apply SRE practices, and when GenAI adds value, I ship retrieval-augmented services (RAG) and lightweight agents exposed via MCP connectors, with quality tracked by offline evaluations and production feedback.

Technical Skills

- Programming: Python, R, SQL
- ML Frameworks: Scikit-learn, TensorFlow, PyTorch, XGBoost
- Data & Orchestration: Apache Spark, Airflow, Kafka, Delta Lake, Feature Stores, MLflow
- Cloud & Infra: Azure, Databricks, Docker, Kubernetes; GCP (Vertex AI, BigQuery, Pub/Sub, Dataflow), AWS; Docker, Kubernetes; Terraform
- MLOps & Ops: MLflow, Model Registry, CI/CD, GitOps, monitoring, observability, Grafana.
- GenAI: Azure OpenAI, RAG, LangChain/LangGraph, agents, MCP, evaluations and guardrails

Professional Experience:

Capgemini Engineering (Client: Royal Caribbean Group) Aug 2024 to Present

Senior Machine Learning Engineer (Tech Lead)

- Led the end-to-end architecture for Propel, a personalization platform that moved from a single-ship pilot to fleet-wide adoption at Celebrity Cruises, driving $5M+ incremental revenue and 10M+ recommendations.
- Evolved the system from batch scoring to event-driven inference with online/offline feature stores, point-in-time correctness, and idempotent pipelines for reliable backfills.
- Built CI/CD with Azure DevOps and Databricks (asset bundles, automated tests) enabling safe retraining and environment promotion; enforced lineage and access policies with Unity Catalog.
- Designed high-concurrency serving and data pipelines with autoscaling, caching and SLO-driven alerting to improve latency and shorten feedback loops; validated high throughput under load.
- Delivered GenAI capabilities: RAG over internal docs/system tables, lightweight agents for operations and MCP connectors to expose internal tools.
- Mentored engineers, set growth plans and aligned roadmap with product and revenue stakeholders.

AB InBev / Bavaria Oct 2021 to Aug 2024

Senior MLOps Engineer

- Technical lead for MLOps in promotion algorithms across 8 countries; integrated models into core processes via Spark/Airflow pipelines and feature stores (batch and streaming).
- Co-built a global ML platform with CI/CD, reproducible training and secure serving; established monitoring, model registry, lineage and role-based access.
- Drove $100M+ incremental revenue by expanding ML algorithms regionally; improved code quality 70%+ through standards and automated testing.

Universidad de los Andes Feb 2024 to Aug 2024

Backend Developer

- Co-designed the backend for a schedule selection platform using FastAPI, Azure Functions, and gRPC; built the middleware to integrate near real-time feeds with the university data bus.
- Supported the teaching team with architecture reviews, code quality and delivery practices.

Comercial Nutresa Feb 2021 to Oct 2021

Data Analyst

- Delivered RPA automations (TagUI, UiPath, VBA, Python) and end-to-end ML models for transport and logistics operations, reducing manual effort and cycle time.

Azimut Energy Jan 2020 to Feb 2021

Data Analyst

- Identified COP $1B+ in energy savings and executed COP $300M in projects; statistical modeling across 1,000+ measurement points; automated reporting for decision support.

Procter & Gamble Jul 2019 to Jan 2020

Project Leader

- Led analytics for utilities, automated data collection and visualization, and evaluated capital projects.

Projects:

Propel: Personalized Offers Engine

- Recommender/ranking workflows with feature stores, experimentation and rollout controls; revenue impact measured with uplift tests.
- High-concurrency serving and SLO-oriented monitoring to sustain peak demand in production.

Global ML Platform

- Training and serving platform with CI/CD, registry and governance; standardized pipelines and observability for multi-country deployments.

GenAI Ops Assistant (RAG & Agents)

- Retrieval over internal docs/system tables; agentic tools and MCP connectors to automate diagnostics and reduce time-to-resolution.

Education

- M.Sc. in Artificial Intelligence, 2023
  Universidad de los Andes
- B.Sc. in Process Engineering, 2019
  Universidad EAFIT

Certifications

Microsoft Azure Data Scientist Associate; Azure Fundamentals; Databricks Platform Administrator; Lakehouse Fundamentals; DS4A Colombia (Honors); IELTS C1

Languages

Spanish (Native), English (C1), German (B1)